Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Department of Computer Technology

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: *To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration*	**Mission:** *To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.*

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

SHEIKH WASIMUDDIN

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Shiky
6/3/26

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Session	**2025-26 (EVEN)**	**Course Name**	**JAVA FSD Lab**
Semester	**6**	**Course Code**	**23IOT1607**
Roll No	**63**	**Name of Student**	**SHEIKH WASIMUDDIN**

Practical Number	**03**
Course Outcome	1. Apply OOP concepts to solve the real-life problem. 2. Use the database concepts to integrate in giving OOP oriented solution. 3. Develop full-stack applications integrating frontend, backend, and database with security. 4. Design and develop mini-project with frontend, backend, and database integration, including authentication and testing.
Aim	Implement the concepts of file handling using the exception mechanism
Problem Definition	1. Write a Java program to append text/ string in a given file 2. Write a Java program to determine number of bytes written to file using Data Output stream. 3.Write a Java program to read content from one file and write it into another file. 4. Write a Java program to get the attributes of file. 5. Read this file and transfer its content in the collection.display the to name of students along with the addition of marks in the three subjects
Theory (100 words)	File handling in Java refers to the process of storing, retrieving, and managing data in files on a computer system. It allows programs to work with external storage so that information can be saved permanently and accessed later. Java provides a number of classes in the java.io and java.nio packages to perform file operations efficiently. Some commonly used classes include File, FileInputStream, FileOutputStream, FileReader, FileWriter, and BufferedReader. The File class is used to create and manage files or directories, while FileReader and BufferedReader help in reading text data from files. Similarly, FileWriter and FileOutputStream are used to write data into files.

	File handling is widely used in applications such as storing user information, generating reports, maintaining logs, and saving configuration data. Java also supports exception handling using IOException to manage errors during file operations, ensuring reliable and secure data processing.
Procedure and Execution (100 Words)	In this practical, different file handling operations are implemented using Java. First, the required classes from the java.io package are imported to perform file operations. A Java program is written to append text into an existing file using the FileWriter class in append mode. Next, DataOutputStream is used to write primitive data types and observe how bytes are written into the file. Another program reads the content from one file using FileReader and copies it into another file using FileWriter. The File class is then used to obtain file attributes such as name, path, and size. Finally, file data is read and stored in a collection to calculate and display student marks. **Algorithm:** 1. Start the program 2. Import necessary classes from java.io and java.util packages. 3. Create or open the required file using the File class. 4. Use FileWriter to append text into the file. 5. Use DataOutputStream to write data and determine bytes written. 6. Read data from one file using FileReader. 7. Write the read content into another file using FileWriter. 8. Retrieve file attributes such as name, path, and size using File methods. 9. Read file data and store it into a collection. 10. Calculate total marks of students and display results. 11. Close all file streams. 12. Stop the program.

Code:

1. Java program to append text/string in a given file.

```
import java.io.FileWriter;
import java.io.IOException;
public class AppendTextFile {
   public static void main(String[] args) {
      try {
         FileWriter fw = new FileWriter("data.txt", true);

         fw.write("\nThis text is appended to the file By Sheikh
Wasimuddin.");
         fw.write("\nJava File Handling Practical Example.");

         fw.close();

         System.out.println("Text appended successfully.");
      }
      catch (IOException e) {
         System.out.println("Error occurred: " + e);
      }
   }
}
```

2. Java program to determine number of bytes written using DataOutputStream.

```
import java.io.DataOutputStream;
import java.io.FileOutputStream;
import java.io.IOException;

public class DataOutputExample {
   public static void main(String[] args) {
      try {
         FileOutputStream fos = new FileOutputStream("byteData.txt");
         DataOutputStream dos = new DataOutputStream(fos);

         dos.writeInt(100);
         dos.writeDouble(55.75);
         dos.writeUTF("Java File Handling By Sheikh Wasimuddin");

         System.out.println("Data written successfully.");

         dos.close();
```

```
            fos.close();
        }
        catch (IOException e) {
            System.out.println(e);
        }
    }
}
```

3. Java program to read content from one file and write it into another file.

```
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;

public class CopyFileContent {
    public static void main(String[] args) {
        try {
            FileReader fr = new FileReader("data.txt");
            FileWriter fw = new FileWriter("destination.txt");

            int ch;

            while ((ch = fr.read()) != -1) {
                fw.write(ch);
            }

            fr.close();
            fw.close();

            System.out.println("File copied successfully.");
        }
        catch (IOException e) {
            System.out.println(e);
        }
    }
}
```

4. Java program to get attributes of a file.

```
import java.io.File;
```

```
public class FileAttributes {
   public static void main(String[] args) {

      File file = new File("data.txt");

      System.out.println("File Name: " + file.getName());
      System.out.println("File Path: " + file.getAbsolutePath());
      System.out.println("File Exists: " + file.exists());
      System.out.println("Readable: " + file.canRead());
      System.out.println("Writable: " + file.canWrite());
      System.out.println("File Size: " + file.length() + " bytes");
   }
}
```

5. Read file and transfer content to collection and display student name with total marks.

```
import java.io.BufferedReader;
import java.io.FileReader;
import java.util.ArrayList;

public class StudentMarks {
   public static void main(String[] args) {
      try {

         BufferedReader br = new BufferedReader(new
FileReader("Student.txt"));
         ArrayList<String> list = new ArrayList<>();

         String line;

         while ((line = br.readLine()) != null) {
            list.add(line);
         }

         for (String s : list) {

            String[] data = s.split(" ");

            String name = data[0];
            int m1 = Integer.parseInt(data[1]);
            int m2 = Integer.parseInt(data[2]);
            int m3 = Integer.parseInt(data[3]);
```

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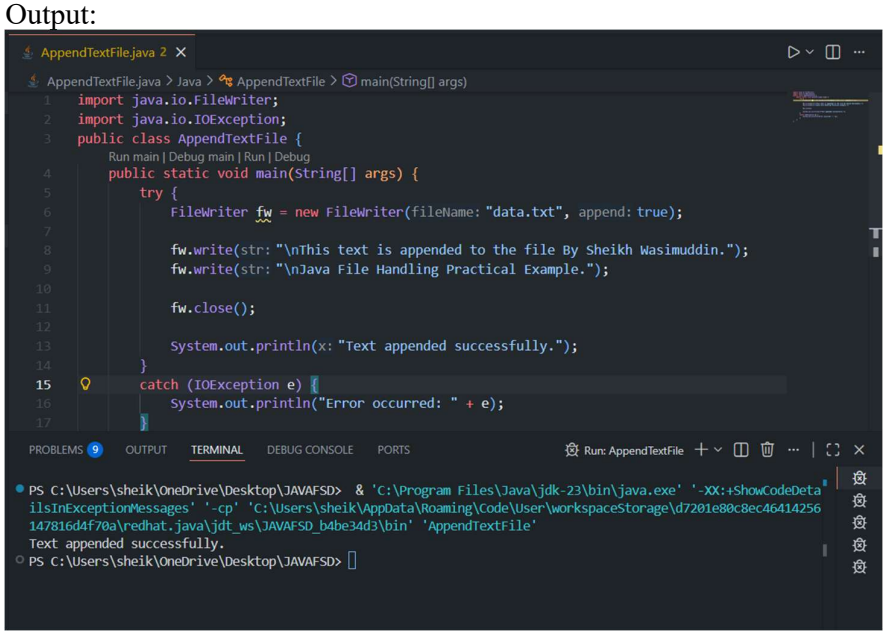
```
            int total = m1 + m2 + m3;

            System.out.println("Name: " + name + " Total Marks: " +
total);
        }

        br.close();
    }
    catch (Exception e) {
        System.out.println(e);
    }
  }
}
```

Output:



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CopyFileContent.java 2

CopyFileContent.java > Language Support for Java(TM) by Red Hat > CopyFileContent > main(String[])

```java
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;

public class CopyFileContent {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) {
        try {
            FileReader fr = new FileReader(fileName: "data.txt");
            FileWriter fw = new FileWriter(fileName: "destination.txt");

            int ch;

            while ((ch = fr.read()) != -1) {
                fw.write(ch);
            }

            fr.close();
```

PROBLEMS 9 OUTPUT TERMINAL DEBUG CONSOLE PORTS Run: CopyFileContent

```
PS C:\Users\sheik\OneDrive\Desktop\JAVAFSD>  & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sheik\AppData\Roaming\Code\User\workspaceStorage\d7201e80c8ec46414256147816d4f70a\redhat.java\jdt_ws\JAVAFSD_b4be34d3\bin' 'CopyFileContent'
File copied successfully.
PS C:\Users\sheik\OneDrive\Desktop\JAVAFSD>
```

DataOutputExample.java 2

DataOutputExample.java > Language Support for Java(TM) by Red Hat > DataOutputExample

```java
import java.io.DataOutputStream;
import java.io.FileOutputStream;
import java.io.IOException;

public class DataOutputExample {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) {
        try {
            FileOutputStream fos = new FileOutputStream(name: "byteData.txt");
            DataOutputStream dos = new DataOutputStream(fos);

            dos.writeInt(v: 100);
            dos.writeDouble(v: 55.75);
            dos.writeUTF(str: "Java File Handling By Sheikh Wasimuddin");

            System.out.println(x: "Data written successfully.");

            dos.close();
```

PROBLEMS 9 OUTPUT TERMINAL DEBUG CONSOLE PORTS Run: DataOutputExample

```
PS C:\Users\sheik\OneDrive\Desktop\JAVAFSD>  & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sheik\AppData\Roaming\Code\User\workspaceStorage\d7201e80c8ec46414256147816d4f70a\redhat.java\jdt_ws\JAVAFSD_b4be34d3\bin' 'DataOutputExample'
Data written successfully.
PS C:\Users\sheik\OneDrive\Desktop\JAVAFSD>
```

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FileAttributes.java

```java
import java.io.File;

public class FileAttributes {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) {

        File file = new File(pathname: "data.txt");

        System.out.println("File Name: " + file.getName());
        System.out.println("File Path: " + file.getAbsolutePath());
        System.out.println("File Exists: " + file.exists());
        System.out.println("Readable: " + file.canRead());
        System.out.println("Writable: " + file.canWrite());
        System.out.println("File Size: " + file.length() + " bytes");
    }
}
```

PROBLEMS 9 OUTPUT TERMINAL DEBUG CONSOLE PORTS Run: FileAttributes

```
PS C:\Users\sheik\OneDrive\Desktop\JAVAFSD>  & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sheik\AppData\Roaming\Code\User\workspaceStorage\d7201e80c8ec46414256147816d4f70a\redhat.java\jdt_ws\JAVAFSD_b4be34d3\bin' 'FileAttributes'
File Name: data.txt
File Path: C:\Users\sheik\OneDrive\Desktop\JAVAFSD\data.txt
File Exists: true
Readable: true
Writable: true
File Size: 185 bytes
PS C:\Users\sheik\OneDrive\Desktop\JAVAFSD>
```

StudentMarks.java 3

```java
import java.io.BufferedReader;
import java.io.FileReader;
import java.util.ArrayList;

public class StudentMarks {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) {
        try {

            BufferedReader br = new BufferedReader(new FileReader(fileName: "Student.txt"));
            ArrayList<String> list = new ArrayList<>();

            String line;

            while ((line = br.readLine()) != null) {
                list.add(line);
            }
```

PROBLEMS 9 OUTPUT TERMINAL DEBUG CONSOLE PORTS Run: StudentMarks

```
PS C:\Users\sheik\OneDrive\Desktop\JAVAFSD>  & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\sheik\AppData\Roaming\Code\User\workspaceStorage\d7201e80c8ec46414256147816d4f70a\redhat.java\jdt_ws\JAVAFSD_b4be34d3\bin' 'StudentMarks'
Name: Sheikh Total Marks: 250
Name: Vedant Total Marks: 252
Name: Tejas Total Marks: 221
Name: Darshil Total Marks: 271
Name: Ketan Total Marks: 218
Name: Tanmay Total Marks: 251
Name: Karan Total Marks: 206
Name: Parth Total Marks: 265
Name: Om Total Marks: 226
Name: Prathamesh Total Marks: 248
PS C:\Users\sheik\OneDrive\Desktop\JAVAFSD>
```

Output Analysis	The given programs demonstrate different file handling operations in Java such as appending data, writing binary data, copying file content, retrieving file attributes, and processing file data. In the first program, text is successfully appended to the data.txt file using the FileWriter class without deleting the existing content, and a confirmation message is displayed. The second program writes different primitive data types like integer, double, and string into byteData.txt using DataOutputStream, confirming successful data storage in the file. The third program reads the content from data.txt and copies it into destination.txt, showing that the file data can be transferred from one file to another. The fourth program retrieves and displays various attributes of data.txt, such as file name, path, existence, permissions, and size. The fifth program reads student records from Student.txt, calculates the total marks for each student, and displays the student name along with their total marks on the console. These outputs confirm that Java file handling allows efficient storage, retrieval, and management of file data.
Link of student Github profile where lab assignment has been uploaded	https://github.com/sheikhwasimuddin
Conclusion	This practical demonstrates the use of file handling in Java to perform various operations such as appending text to a file, writing data using DataOutputStream, copying content from one file to another, retrieving file attributes, and reading data to process student marks. The programs show how Java classes from the java.io package help in managing files efficiently. By using these classes, data can be stored, accessed, and modified easily. Proper use of exception handling ensures that errors during file operations are handled safely. Overall, file handling is an important concept in Java that allows programs to store and manage data permanently in an organized manner.

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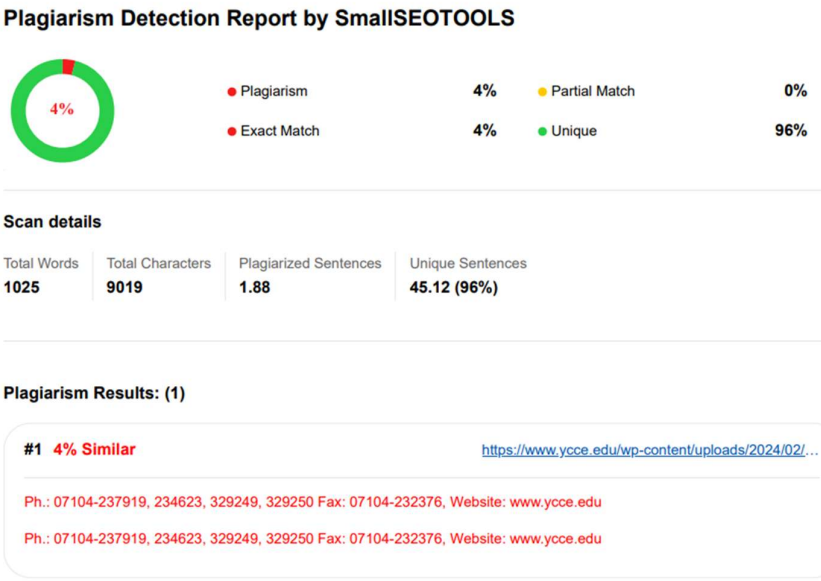
Plag Report (Similarity index < 12%)	 **Plagiarism Detection Report by SmallSEOTOOLS** 4% Plagiarism 4% Partial Match 0% Exact Match 4% Unique 96% **Scan details** Total Words 1025 | Total Characters 9019 | Plagiarized Sentences 1.88 | Unique Sentences 45.12 (96%) **Plagiarism Results: (1)** #1 4% Similar https://www.ycce.edu/wp-content/uploads/2024/02/... Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu
Date	**06/03/2026**

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